3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius r, as shown in Fig. 3.1.

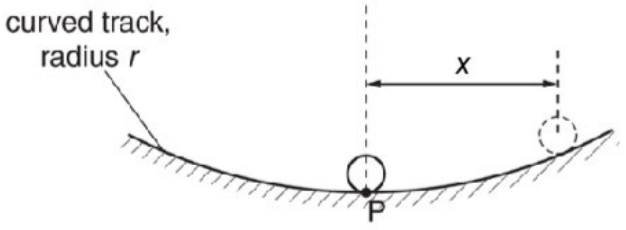


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement x of the ball is related to its acceleration a towards P by the expression

$$a = -\frac{gx}{r}$$

where g is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

.....

.....

.....

.....[2]

(ii) The radius r of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

τ = ...s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

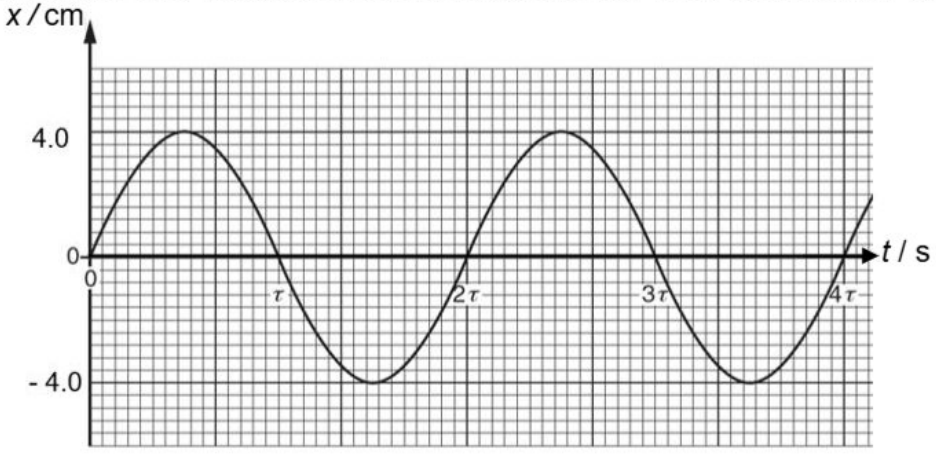


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

maximum kinetic energy= .....J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

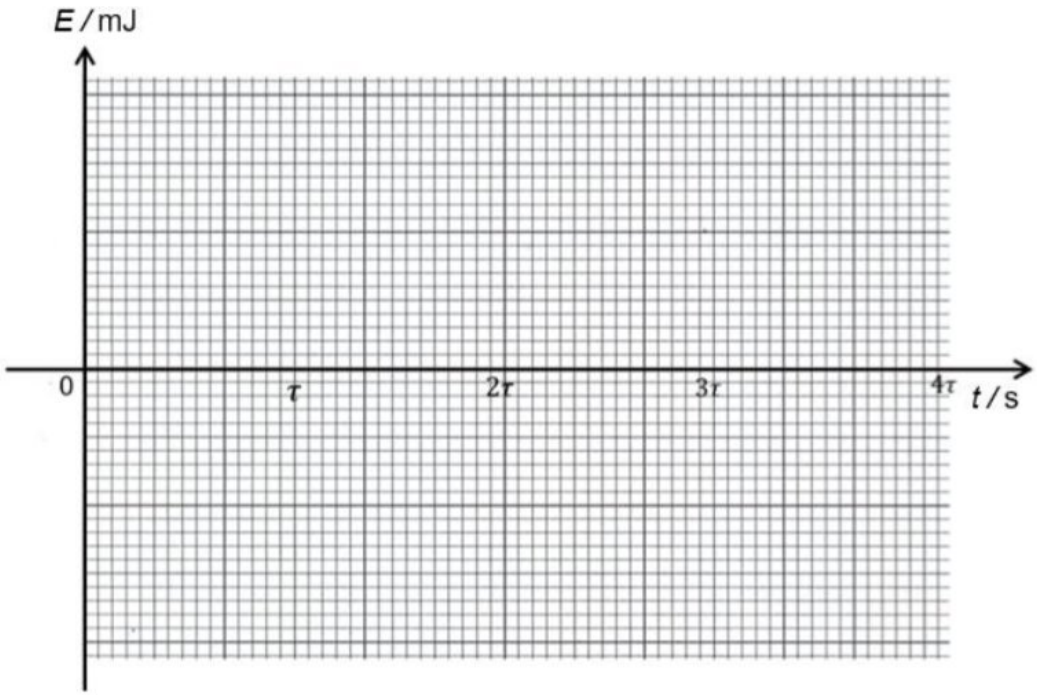


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

[Total: 9]